Oscar **Poudel**

Ph.D. in Civil Engineering
Specialization in AI and Robotics for Construction Automation

https://oscarp.com.np
linkedin.com/in/oscar-poudel
github.com/oscarpoudel
https://scholar.google.com/citations?user=NWVoANsAAAAJhl=enauthuser=1
+201 726 0613
op72@njit.edu
Newark, NJ, 07102

Research Interests

- **Human-Robot Interaction for Construction Robotics :** Designing and implementing hardware-software systems to facilitate multimodal interactions between humans and robots, with applications in construction site automation and assistive robotics.
- **Structural Health Monitoring and Inspection :** Development of computer vision–based systems for automated detection, quantification, and analysis of structural conditions, utilizing camera sensors, edge computing, and embedded AI for real-time infrastructure assessment.
- **Artificial Intelligence for Civil and Construction Engineering :** Application of large language models (LLMs), deep learning (DL), and reinforcement learning (RL) for construction automation, predictive modeling, and intelligent decision-making, including the design of agentic systems and autonomous workflows.
- **Intelligent Infrastructure and Ubiquitous Monitoring Systems :** Integration of ubiquitous sensing technologies, wireless sensor networks, and data fusion techniques to enable scalable, adaptive monitoring and predictive maintenance for civil infrastructure.

Professional Experience

Present – September 2023
Research Assistant | AI and Automation Engineer, New Jersey Institute of Technology (NJIT), Newark, NJ, USA

- Developed human-robot interaction systems for autonomous construction tasks using ROS2, Isaac Sim, and embedded systems
- Integrated large language models (LLMs), computer vision, and reinforcement learning for construction site automation
- Led the development of an agent-based simulation platform for infrastructure maintenance planning
- Contributed to grant writing and conference papers in robotics, AI, and civil engineering domains

Python | PyTorch | ROS2 | Isaac Sim | Gazebo | LangChain | PydanticAI | Git | Arduinos | Microcontroller Programming

Summer 2024 & Summer 2025
Construction AI Engineer, UrbanTech Consulting Engineering, New York, NY 10004

- Designed and trained transformer-based deep learning models for infrastructure inspection and defect analysis
- Developed multimodal image–LiDAR fusion pipelines for 3D crack segmentation in structural models
- Implemented 3D computer vision workflows for defect detection on mesh and point cloud data
- Integrated trained deep learning models into a user-facing GUI/application to enable practical deployment within the company
- Conducted applied research on advanced segmentation and data fusion techniques to improve model accuracy and robustness

Pytorch | ONNX | Deep Learning | User Interface | Computer Vision

September 2025 – September 2023
Teaching Assistant | Civil Engineering Department, NJIT, Newark, NJ, USA

- Assisted in teaching undergraduate courses including Strength of Materials, Statistics, and Engineering Dynamics
- Conducted recitation sessions, developed grading rubrics, and supported student engagement and mentoring
- Coordinated lab assignments and provided technical support during simulation-based instruction

SolidWorks | Excel | LaTeX

August 2023 – November 2022
Structural Engineer, Hatemalo Engineering & Construction Pvt. Ltd., Kathmandu, Nepal

- Performed structural analysis and design of RCC and steel buildings including seismic detailing per NBC/IS codes
- Conducted site inspections and verification of construction practices aligned with design documents
- Collaborated with architectural and MEP teams for coordinated BIM workflows

ETABS | SAP2000 | Rhino | AutoCAD | Revit | Excel | SketchUp

November 2022 | **Junior Structural Engineer, MRB & Associates, Kathmandu, Nepal**
August 2022

- Supported senior engineers in analysis and detailing of multi-storey commercial and residential structures
- Participated in retrofitting design and strengthening evaluation of vulnerable RC buildings
- Prepared structural drawings, BOQs, and engineering documentation

ETABS | SAP | SAFE | AutoCAD | Excel

Education

2023-Present	Ph.D. in Civil Engineering (Construction Automation/ Construction Management), New Jersey Institute of Technology, Newark, NJ
2023-Present	Masters in Artificial Intelligence (MSAI), New Jersey Institute of Technology, Newark, NJ
2017-2022	Bachelor's in Civil Engineering, Institute of Engineering, Pulchowk Campus, Lalitpur, Nepal

Technical Skills

Civil Engineering	**ETABS**, **SAP2000**, **SAFE**, **PERFORM3D**, Robot Structural Analysis, RCDC, IDEA StatiCa, OPENSEES, Grasshopper
Drafting & BIM Tools	**Revit**, **AutoCAD**, SketchUp, Lumion
GIS & Remote Sensing	**QGIS**, SNAP (ESA), Civil3D
Project Management Tools	**Primavera P6**, **MS Project**
Programming Languages	**Python**, **C/C++**, **MATLAB**, JavaScript
AI & GenAI Frameworks	**PyTorch**, **TensorFlow**, **LangChain**, **PydanticAI**, scikit-learn, OpenCV
Simulation & Robotics Platforms	**ROS2**, **Isaac Sim**, **Webots**, Unity AR/VR, Gazebo, MoveIt, Docker, AWS
Embedded Systems & Hardware	**Arduino**,**PCB design**, IOTs(NodeRed,C++), Microcontrollers, Sensor Integration
Professional Skills	Scientific Reporting, Grant Writing, Technical Presentation, Teaching and Mentoring
Operating Systems	**Linux**, **Windows**, macOS

Certification and Licensure

2026	NCEES PE Civil Structural (2571691-1979841-5b8c515)
2026	Chi-Epsilon, NJIT Chapter (135749)
2025	NCEES FE Civil (2571691-1865513-e1a7b26)
2023	Nepal Engineering Council (Category "A" 32803)
2022	ASCE member (000012454689)

Research Publicaitons

Journal Papers (J)

J08. **Poudel, O.**, Assaad, R. H., Park, A. B., & Awada, M. (2027; published online 2026). MAESTRO : Fine-Tuned Multiagent Large Action Model-Based GenAI Assistant Integrating IoT Sensing, Smart Control, and Extended Reality for Human-Building Interaction. *Journal of Computing in Civil Engineering*, 41(1).

J07. Park, A. B., **Poudel, O.**, Wu, Z., Awada, M., & Assaad, R. H. (2026). Age-group perspectives on large language models in the architecture, engineering, and construction industry : Usage patterns, adoption expectations, trust, and future outlook. *Construction Management and Economics*, 1–19.

J06. **Poudel, O.**, Assaad, R. H., & Awada, M. (2026). Noise-Invariant Agentic Human-Robot Interaction GenAI System Using a Dual-Encoder Contrastive ASR Architecture and VLMs for Robot Control and Navigation in Acoustically Challenging Jobsites. *Journal of Computing in Civil Engineering*, 40(6), 04026088.

J05. **Poudel, O.**, Mohammadi, M., Assaf, G., & Assaad, R. H. (2026). Intelligent Acoustic-Based IoT-Enabled Sensing and Monitoring System for Automated Real-Time Leak Detection, Localization, and Pinpointing in Water Distribution Infrastructures. *Journal of Infrastructure Systems*, 32(3), 04026017.

J04. **Poudel, O.**, Hu, X., Assaad, R. H., Wang, W., Huang, T., & Wang, G. (2026). VTECSeg : Edge-Aware Hybrid CNN-Vision Transformer Network with Zero-Shot Vision-Language-Guided Region Proposals for Crack Segmentation. *Automation in Construction*, 188, 106998.

J03. **Poudel, O.**, & Assaad, R. H. (2025). A Real-Time Intelligent Acoustic IoT-Enabled Embedded Construction Site Monitoring and Alert System : Integrating Deep Learning-Based Machine-Listening Algorithms, Edge Computing, and Cloud Computing. *Journal of Construction Engineering and Management*, 151(7), 04025075.

J02. Assaad, R. H., Mohammadi, M., & **Poudel, O.** (2025). Developing an Intelligent IoT-Enabled Wearable Multimodal Biosensing Device and Cloud-Based Digital Dashboard for Real-Time and Comprehensive Health, Physiological, Emotional, and Cognitive Monitoring Using Multi-Sensor Fusion Technologies. *Sensors and Actuators A : Physical*, 381, 116074.

J01. **Poudel, O.**, & Shrestha, K. C. (2024). Enhancing Earthquake Resilience with Strategically Arranged Inclined Columns in Mul-

tistoried RCC Structures. *Innovative Infrastructure Solutions*, 9(6), 223.

Conference Papers (C)

C11. Charbel, G., **Poudel, O.**, Assaf, M., Assaad, R. H., & Awada, M. (2026). Pavement Deterioration Prediction by Integrating Agent-Based Simulation and Transformer-Based Machine Learning Models for Smart Infrastructure Asset Management. In *Construction Research Congress 2026*, 502–512.

C10. **Poudel, O.**, Charbel, G., Assaad, R. H., & Awada, M. (2026). BIM Augmented VR Teleoperation for Precise Robotic Arm Manipulation with Human-in-the-Loop in Low Level Construction Tasks. In *Construction Research Congress 2026*, 100–109.

C09. **Poudel, O.**, Assaad, R. H., & Awada, M. (2026). AI-Assisted Autonomous Quadruped Mobile Robotic System Integrated with Thermal and Environmental Sensing for Real-Time Fire Hazard Detection and Mitigation in Construction and Facility Safety Management. In *Construction Research Congress 2026*, 90–99.

C08. **Poudel, O.**, Assaf, M., Assaad, R. H., & Awada, M. (2026). Talk, See, Support (TSS) : A Wearable Multimodal, Intelligent, Agentic LLM-Powered Construction Assistant for Real-Time On-Site Guidance and Compliance Monitoring. In *Construction Research Congress 2026*, 94–104.

C07. **Poudel, O.**, Assaad, R. H., & Awada, M. (2025). Viso-Gen : A Wireless Gaze-Assisted and Voice-Controlled Generative AI System Leveraging Stable Diffusion for Real-Time Image Generation for Civil Engineering Design and Visualization. In *Computing in Civil Engineering 2025*, 990–999.

C06. **Poudel, O.**, Assaad, R. H., & Awada, M. (2025). Integrating Large Language Models (LLMs) with Autonomous Aerial Drone Robotics and Computer Vision for Contextual Adaptive Construction Site Safety Management and Risk Assessment. In *Computing in Civil Engineering 2025*, 509–518.

C05. **Poudel, O.**, Assaad, R. H., & Awada, M. (2025). SLAM-Assisted Transformer-Based TD3 Deep Reinforcement Learning for Adaptive Navigation and Mapping and Dynamic Obstacle Avoidance in Autonomous Construction Robotics. In *Computing in Civil Engineering 2025*, 519–529.

C04. Mohammadi, M., **Poudel, O.**, Assaad, R. H., Awada, M., & Assaf, G. (2025). A Smart Cyber-Physical System Combining Large Language Models (LLMs) and Augmented Reality (AR) to Enhance Human-Building Interaction during Building Operations and Maintenance Procedures. In *Computing in Civil Engineering 2025*, 847–857.

C03. Mohammadi, M., **Poudel, O.**, Assaad, R. H., Awada, M., & Assaf, G. (2025). An Intelligent Cyber-Physical System Integrating Large Language Models (LLMs), IoT Wireless Sensor Networks, and Augmented Reality (AR) for Real-Time Indoor Environmental Quality Improvement and Automated HVAC Control in Smart Buildings. In *Computing in Civil Engineering 2025*, 980–989.

C02. **Poudel, O.**, & Assaad, R. H. (2024). Developing Real-Time Acoustic Technologies for Enhancing Safety and Efficiency on Construction Job Sites Using Deep Learning Algorithms : Moving Towards Automated Construction Site Monitoring. In *Computing in Civil Engineering 2024*, 1009–1018.

C01. **Poudel, O.**, & Assaad, R. H. (2024). Advancing Sound Level Localization through Innovative Integration and Fusion of Remote Sensing and Reality Captured LiDAR-Based Digital Twins, Auditory Devices, and IoT-Enabled Sensing Technologies. In *Computing in Civil Engineering 2024*, 466–475.